

YAZAN SALLOUM

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PROFESSIONAL SUMMARY

I'm a third-year engineering student predicted to graduate with First Class Honours, with three years of Formula Student experience, two of which were spent as Powertrain Lead. I delivered simulation-driven component design under tight packaging constraints and competition deadlines. I am CSWA-certified in SOLIDWORKS, and have strong CFD/FEA and rapid-prototyping skills (Ansys, StarCCM+, FDM), along with hands-on instrumentation experience.

TECHNICAL SKILLS

CAD & Design: SOLIDWORKS (CSWA certified), Autodesk Fusion, GD&T, design for manufacture (DFM), rapid prototyping, FDM 3D printing
Simulation & Analysis: Ansys Fluent, Zemax OpticStudio, Siemens StarCCM+, computational fluid dynamics (CFD), finite element analysis (FEA)
Programming & Tools: Python, MATLAB, Simulink, C++, Microsoft Excel, PowerPoint, Word
Engineering Methods: Design documentation, Bill of Materials (BoM/CBoM), Gantt-based project planning, instrumentation & measurement, failure-mode analysis, Granta EduPack material selection
Certifications: [SOLIDWORKS Associate \(CSWA\)](#); [IBM Enterprise Design Thinking Practitioner](#)
Languages: Arabic (native), English (native level), French (intermediate), German (beginner), Spanish (beginner)

ENGINEERING EXPERIENCE

Queen Mary Formula Student

Sep 2023 – Present

Powertrain Lead

Jul 2024 – Present

- Leading development of a digital twin for the 2026 FSUK entry, to be used as a driver-familiarisation tool to improve performance in dynamic events.
- Ran static engine testing under controlled lab conditions, acquiring and analysing sensor data.
- Designed a CFD-optimised 4-1 exhaust header and collector around tight packaging constraints.
- Restructured the powertrain department, increasing team size and output by 200%.
- Managed design and documentation timelines via Gantt charts, reducing development cycle time by 40%.
- Created standard operating procedures and safety protocols to reduce workshop hazards.
- Created the majority of departmental documentation, including BoM sheets and design reports.
- Key contributor to the team's overall 6th-place finish at FSUK 2025 (Concept).

Senior Powertrain Engineer

Sep 2023 – Jun 2024

- Designed the intake system for the 2024 Concept entry using iterative CFD simulations to refine performance.
- Strengthened research skills by analysing external design strategies to improve own design.
- Helped the team secure a 6th-place finish at FSUK 2024 (Concept).

Continuous Advanced Recording for Cancer Lab — UG Researcher

Sep 2025 – Present

- Supporting PhD and postdoctoral researchers on bioelectronic systems for cancer monitoring.

EDUCATION

Queen Mary University of London — BEng (Hons)

Sep 2023 – Present

Biomedical Engineering – predicted classification: First Class Honours (1:1)

Third Year Project — Design & Manufacture of a Compact Mobile Phone Spectrometer Attachment

- Led the housing and layout workstream for a sub-£200 portable smartphone spectrometer (a low-cost optical diagnostic device), using Autodesk Fusion for CAD, Ansys Zemax OpticStudio for optical simulation, and Granta EduPack for material selection.

- Created multiple prototypes to validate design via bench testing, with the finalised version meeting three of four targets: £182.21 unit cost, 70g mass (150g target), and 0.18 nm/pixel spectral resolution, surpassing published literature.

David Game College London — University Foundation Programme

Jan 2023 – Aug 2023

- A*A*A in Further Mathematics, Further Computing, and Physics

Earlier qualifications:

- Syrian Scientific Baccalaureate: 98.2% (2356/2400).
- SAT 1480/1600, including 800/800 in Mathematics.
- Syrian Scientific Mathematics Olympiad 2019 — ranked 17th nationally.

SELECTED PROJECTS

- **Femoral Fracture Fixation Device:** Led CAD, material selection, and FEA of an intramedullary nail design to be used for femoral fracture fixation, balancing feasibility with clinical performance.
- **EMG Signal Analysis:** Captured and processed EMG test data in MATLAB (bandpass filtering, rectification, RMS windowing), comparing muscle-activation phases across subjects.

ADDITIONAL EXPERIENCE

Tom Brown at The Capital — Food & Beverage Assistant

Apr 2026 – Present

- Delivering high-standard service at a Michelin-starred restaurant alongside full-time study.

Queen Mary Centre of Undergraduate Research — UG Researcher

May 2024 – Mar 2025

- Ran a data-collection study on the effects of ageing societies on the United Kingdom's policymaking, healthcare, technology, and accident rates.
- Presented findings at QMUL's Industrial Liaison Forum to an audience of 100+ faculty and industry professionals.

Queen Mary Petrolhead Society — President

Apr 2025 – Jun 2026

- Led committee operations and event programming, resulting in growing membership numbers, building on prior term as Vice President.